NTIRE 2026 Image Super-Resolution (×4) Challenge Factsheet Restore Fidelity First, then Generate Details for Perceptual Image Super-Resolution

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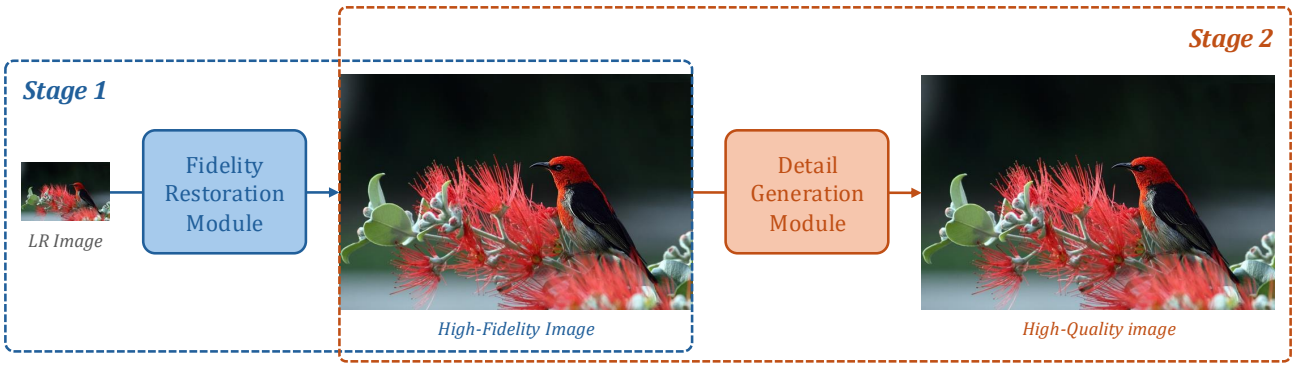

Figure 1. Overview of the two-phase LoRA training strategy used in the detail generation module. Left) In the first phase, we train a LoRA with the flow-matching loss for faithful restoration. Right) In the second phase, we freeze the first-phase LoRA and train an additional LoRA to improve perceptual quality by minimizing LPIPS and maximizing MUSIQ.

1. Factsheet Information

1.1. Team Details

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- Link to the codes/executables of the solution(s): github repository

2. Method Details

2.1. Overview

Recent diffusion-based super-resolution (SR) models [5, 16, 24, 28, 31] have shown impressive results by generating realistic details using the rich priors of text-to-image generation models [6, 20, 21]. However, they often struggle to maintain fidelity and structural consistency with respect to the low-resolution (LR) input [2, 10, 18]. To maximize fidelity while still enabling realistic detail generation, we adopt a two-stage pipeline inspired by DiffBIR [16]: restore fidelity first, and then generate realistic details. An overview of our pipeline is shown in Figure 1.

2.2. Stage 1: Fidelity Restoration Module

We considered several candidates for the fidelity restoration module and evaluated them on the DIV2K [1] validation set. Among them, HAT [3] achieved the highest PSNR, specifically with the HAT-L_SRx4_ImageNet-pretrain model. Therefore, we adopt HAT as the first-stage module without further fine-tuning.

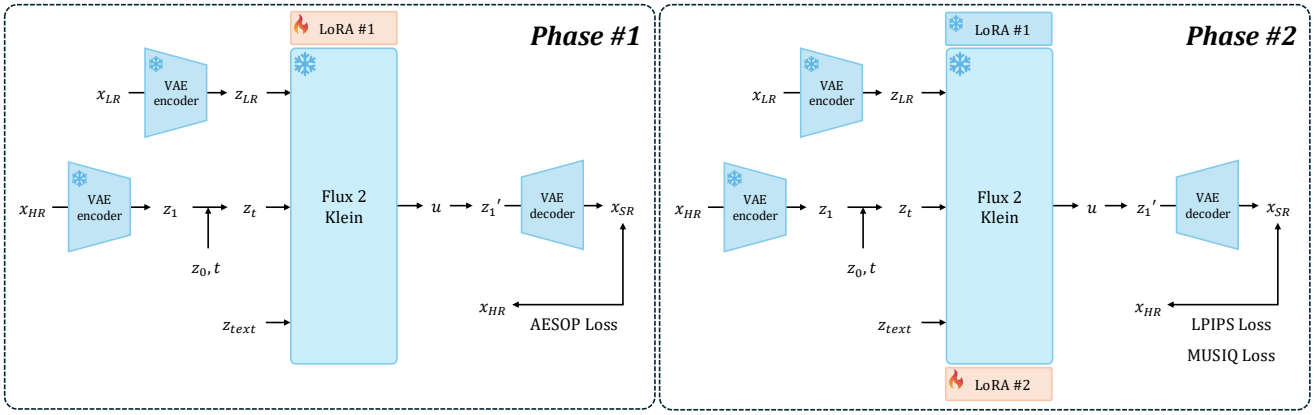

Figure 2. Overview of the training pipeline of our method. Left) We first train a LoRA module with the AESOP loss for faithful restoration. Right) We then train a second LoRA on top of it to further enhance perceptual quality by minimizing LPIPS and maximizing MUSIQ.

2.3. Stage 2: Detail Generation Module

Previous diffusion-based SR models [2, 5, 16, 22, 24, 28, 31, 32] have primarily used text-to-image generation models [6, 20, 21] as their backbones. A key challenge in these approaches is how to inject LR image information while preserving both fidelity and realistic details. Although many methods have been proposed, appropriately incorporating LR information remains far from trivial.

Recently, foundation image editing models that use text instructions [12, 13, 26] have demonstrated impressive performance. Motivated by these advances, we conduct preliminary prompt-based experiments to evaluate their potential for SR. We find that some of these models can generate realistic details while preserving fidelity to a certain extent, even without additional training. This observation suggests that foundation image editing models can serve as a promising starting point for SR, compared with conventional text-to-image generation models.

However, directly applying foundation image editing models to SR presents two major challenges. First, their output fidelity to the LR input remains insufficient. For example, we observe noticeable changes in color tone and structural inconsistencies, such as altered cloud shapes. Second, these models cannot readily handle high-resolution inputs, as they implicitly resize the conditioning image when the input resolution is large. Simply removing this resize operation leads to subpar results. To address these issues, we employ LoRA [9] training for the first challenge and sliding-window inference for the second. Considering our computational budget and preliminary qualitative comparisons, we use FLUX 2 Klein [12] (FLUX 2 Klein base 9B) as the backbone of the detail generation module.

2.3.1. LoRA Training

Various SR models [14, 22, 27, 32, 33] have improved performance by incorporating additional training losses [7, 8, 11, 15, 25, 30, 34]. However, the choice of loss combinations varies widely across studies, and there has been little systematic investigation into which combinations are most effective. To better understand this design space, we conducted extensive experiments with diverse loss combinations [4, 11, 14, 17, 19, 23, 29, 34] and obtained four key observations. 1) Some losses exhibit strong synergy, whereas others do not. 2) Proper weight balancing between losses is essential. 3) Sequential optimization, as also observed in PiSA-SR [22], is more effective than conventional joint optimization. 4) In sequential optimization, the order of losses is crucial.

Motivated by these findings, we adopt a two-phase LoRA training strategy for the detail generation module. In the first phase, we train a LoRA with the AESOP loss [14], which measures the difference between generated and target images in the output space of a pre-trained autoencoder rather than in raw pixel space, to obtain a fidelity-preserving adaptation. In the second phase, we freeze the first-phase LoRA and train an additional LoRA on top of it for perceptual enhancement. Specifically, we optimize a weighted combination of LPIPS [34] and MUSIQ [11]-based objectives, i.e., $L_{LPIPS} + 0.35 \times L_{MUSIQ} = LPIPS + 0.35 \times (1 - MUSIQ)$. The overall training pipeline is illustrated in Figure 2.

2.3.2. Sliding Window Inference

Foundation image editing models cannot readily handle arbitrary high-resolution conditioning inputs. Directly using high-resolution images often leads to suboptimal results. Moreover, simply dividing the image into overlapping patches and stitching the outputs is not well suited for diffusion-based generation, because discrepancies between patches are accumulated over the denoising process. To address this issue, we follow the patch aggregation sampling strategy widely used in diffusion-based SR meth-

ods [16, 24, 31, 32], where overlapping patches are processed independently and their predictions are fused using Gaussian weights at each denoising step.

2.4. Hyperparameters

2.4.1. Train Details

- Dataset: DIV2K train
- Data Augmentation: 1024 x 1024 random crop
- Text prompt: "Use the upscaled condition image as a strict reference. Perform only faithful x4 restoration refinement: reduce blur, ringing, aliasing, and compression artifacts, and recover natural fine details only where supported by the input. Keep the exact same scene layout, object shapes, colors, lighting, and semantics. Improve edge clarity and local texture realism without hallucinating new structures, oversharpening, or changing content."
- LoRA target modules: only to the q, k, v, and out projections in the image stream of the rectified flow transformer
- LoRA rank, alpha: 128, 256
- Iteration: 2000
- Batch size: 2
- Optimizer: AdamW
- Learning rate: 0.000095
- Weight decay: 0.0001

2.4.2. Inference Details

- Text prompt: same text prompt used in train
- Guidance scale: 3.5
- Inference steps: 50
- Patch size: 1024 x 1024
- Patch overlap: 512 x 512

3. Other Details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop: Perhaps yes.
- General comments and impressions of the NTIRE 2026 challenge: We sincerely appreciate the organizers' hard work for preparing the NTIRE 2026 challenge and providing such a valuable research opportunity for the community.
- What do you expect from a new challenge in image restoration, enhancement and manipulation: Real-world image super-resolution.
- Other comments: For Track 2, which evaluates perceptual quality, the scoring appears to be biased toward NR-IQA metrics. We believe it would be fairer to balance the evaluation more evenly between full-reference IQA metrics and no-reference IQA metrics, for example with a 1:1 ratio: $2 \times ((1 - LPIPS) + (1 - DISTS)) + ((1 - NIQE/10) + ManIQA + MUSIQ/100 + CLIP - IQA)$.

References

[1] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 126–135, 2017. 1

[2] Junyang Chen, Jinshan Pan, and Jiangxin Dong. Faithdiff: Unleashing diffusion priors for faithful image super-resolution. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 28188–28197, 2025. 1, 2

[3] Xiangyu Chen, Xintao Wang, Jiantao Zhou, Yu Qiao, and Chao Dong. Activating more pixels in image super-resolution transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22367–22377, 2023. 1

[4] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P Simoncelli. Image quality assessment: Unifying structure and texture similarity. *IEEE transactions on pattern analysis and machine intelligence*, 44(5):2567–2581, 2020. 2

[5] Zheng-Peng Duan, Jiawei Zhang, Xin Jin, Ziheng Zhang, Zheng Xiong, Dongqing Zou, Jimmy S Ren, Chunle Guo, and Chongyi Li. Dit4sr: Taming diffusion transformer for real-world image super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 18948–18958, 2025. 1, 2

[6] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024. 1, 2

[7] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014. 2

[8] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022. 2

[9] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022. 2

[10] Qiming Hu, Linlong Fan, Yiyan Luo, Yuhang Yu, Xiaojie Guo, and Qingnan Fan. Text-aware real-world image super-resolution via diffusion model with joint segmentation decoders. *arXiv preprint arXiv:2506.04641*, 2025. 1

[11] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5148–5157, 2021. 2

[12] Black Forest Labs. FLUX.2: Frontier Visual Intelligence. https://bfl.ai/blog/flux-2, 2025. 2

[13] Black Forest Labs, Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, et al. Flux. 1 kontext: Flow matching for in-context image generation and editing in latent space. *arXiv preprint arXiv:2506.15742*, 2025. 2

[14] MinKyu Lee, Sangeek Hyun, Woojin Jun, and Jae-Pil Heo. Auto-encoded supervision for perceptual image super-resolution. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 17958–17968, 2025. 2

[15] Jie Liang, Hui Zeng, and Lei Zhang. Details or artifacts: A locally discriminative learning approach to realistic image super-resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5657–5666, 2022. 2

[16] Xinqi Lin, Jingwen He, Ziyan Chen, Zhaoyang Lyu, Bo Dai, Fanghua Yu, Yu Qiao, Wanli Ouyang, and Chao Dong. Diffbir: Toward blind image restoration with generative diffusion prior. In *European conference on computer vision*, pages 430–448. Springer, 2024. 1, 2, 3

[17] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022. 2

[18] Jaewon Min, Jin Hyeon Kim, Paul Hyunbin Cho, Jaeeun Lee, Jihye Park, Minkyu Park, Sangpil Kim, Hyunhee Park, and Seungryong Kim. Text-aware image restoration with diffusion models. *arXiv preprint arXiv:2506.09993*, 2025. 1

[19] Anish Mittal, Rajiv Soundararajan, and Alan C Bovik. Making a "completely blind" image quality analyzer. *IEEE Signal processing letters*, 20(3):209–212, 2012. 2

[20] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023. 1, 2

[21] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 1, 2

[22] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. Pixel-level and semantic-level adjustable super-resolution: A dual-lora approach. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2333–2343, 2025. 2

[23] Jianyi Wang, Kelvin CK Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *Proceedings of the AAAI conference on artificial intelligence*, pages 2555–2563, 2023. 2

[24] Jianyi Wang, Zongsheng Yue, Shangchen Zhou, Kelvin CK Chan, and Chen Change Loy. Exploiting diffusion prior for real-world image super-resolution. *International Journal of Computer Vision*, 132(12):5929–5949, 2024. 1, 2, 3

[25] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolificdreamer: High-fidelity and diverse text-to-3d generation with variational score distillation. *Advances in neural information processing systems*, 36: 8406–8441, 2023. 2

[26] Chenfei Wu, Jiahao Li, Jingren Zhou, Junyang Lin, Kaiyuan Gao, Kun Yan, Sheng ming Yin, Shuai Bai, Xiao Xu, Yilei Chen, Yuxiang Chen, Zecheng Tang, Zekai Zhang, Zhengyi Wang, An Yang, Bowen Yu, Chen Cheng, Dayiheng Liu, Deqing Li, Hang Zhang, Hao Meng, Hu Wei, Jingyuan Ni, Kai Chen, Kuan Cao, Liang Peng, Lin Qu, Minggang Wu, Peng Wang, Shuting Yu, Tingkun Wen, Wensen Feng, Xiaoxiao Xu, Yi Wang, Yichang Zhang, Yongqiang Zhu, Yujia Wu, Yuxuan Cai, and Zenan Liu. Qwen-image technical report, 2025. 2

[27] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. One-step effective diffusion network for real-world image super-resolution. *Advances in Neural Information Processing Systems*, 37:92529–92553, 2024. 2

[28] Rongyuan Wu, Tao Yang, Lingchen Sun, Zhengqiang Zhang, Shuai Li, and Lei Zhang. Seesr: Towards semantics-aware real-world image super-resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 25456–25467, 2024. 1, 2

[29] Sidi Yang, Tianhe Wu, Shuwei Shi, Shanshan Lao, Yuan Gong, Mingdeng Cao, Jiahao Wang, and Yujiu Yang. Maniqa: Multi-dimension attention network for no-reference image quality assessment. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1191–1200, 2022. 2

[30] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Fredo Durand, William T Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6613–6623, 2024. 2

[31] Fanghua Yu, Jinjin Gu, Zheyuan Li, Jinfan Hu, Xiangtao Kong, Xintao Wang, Jingwen He, Yu Qiao, and Chao Dong. Scaling up to excellence: Practicing model scaling for photo-realistic image restoration in the wild. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 25669–25680, 2024. 1, 2, 3

[32] Zongsheng Yue, Kang Liao, and Chen Change Loy. Arbitrary-steps image super-resolution via diffusion inversion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 23153–23163, 2025. 2, 3

[33] Fengjia Zhang, Samrudhdhi B Rangrej, Tristan Aumentado-Armstrong, Afsaneh Fazly, and Alex Levinshtein. Augmenting perceptual super-resolution via image quality predictors. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 2311–2322, 2025. 2

[34] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. 2